

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.24681612749809353, median 0.21766761528510062, std: 0.15314870761679272
Reprojection error (cam1): mean 0.24469093923620566, median 0.21129106983295856, std: 0.1609817126466607
Gyroscope error (imu0): mean 0.2111356726505709, median 0.18284137928247576, std: 0.16322343646305662
Accelerometer error (imu0): mean 0.2606629998881611, median 0.21823035892895898, std: 0.18613316656303625

Residuals

Reprojection error (cam0) [px]: mean 0.24681612749809353, median 0.21766761528510062, std: 0.15314870761679272
Reprojection error (cam1) [px]: mean 0.24469093923620566, median 0.21129106983295856, std: 0.1609817126466607
Gyroscope error (imu0) [rad/s]: mean 0.007722095938047016, median 0.006687257792769308, std: 0.00596974930802645
Accelerometer error (imu0) [m/s^2]: mean 0.0851894813825314, median 0.07132171081840628, std: 0.060831755693737694

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[-0.01455298 -0.99982739 -0.0115501  0.01073148]
 [-0.01583116  0.01178027 -0.99980528 -0.04255252]
 [ 0.99976877 -0.01436729 -0.01599986 -0.11677268]
 [ 0.          0.          0.          1.          ]]
```

T_ic: (cam0 to imu0):

```
[[-0.01455298 -0.01583116  0.99976877  0.1162282 ]
 [-0.99982739  0.01178027 -0.01436729  0.0095532 ]
 [-0.0115501  -0.99980528 -0.01599986 -0.04428863]
 [ 0.          0.          0.          1.          ]]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)

0.0016570249687064458

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.00419894 -0.99992567 -0.01144694 -0.04112276]
[-0.00787567 0.01141362 -0.99990385 -0.04357658]
[0.99996017 0.00428868 -0.00782715 -0.11572966]
[0. 0. 0. 1.]]

T_ic: (cam1 to imu0):
[[0.00419894 -0.00787567 0.99996017 0.11555452]
[-0.99992567 0.01141362 0.00428868 -0.040126]
[-0.01144694 -0.99990385 -0.00782715 -0.04494895]
[0. 0. 0. 1.]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.0016540230686333628

Baselines:

Baseline (cam0 to cam1):
[[0.99982417 -0.00040116 0.01874734 -0.04968024]
[0.00025195 0.99996828 0.0079605 -0.00009855]
[-0.01874994 -0.00795437 0.99979256 0.00088154]
[0. 0. 0. 1.]]
baseline norm: 0.049688156791381215 [m]

Gravity vector in target coords: [m/s^2]
[-9.80346365 0.07709915 -0.23362147]

Calibration configuration
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cam0

Camera model: pinhole
Focal length: [382.7017442149056, 378.72007312002575]
Principal point: [322.56276537323123, 248.49645193348633]
Distortion model: radtan
Distortion coefficients: [0.046438306688819736, -0.07604899763746233, 0.0006120080138400965, 0.004310326482994855]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.02 [m]
Spacing 0.006 [m]

cam1

Camera model: pinhole
Focal length: [384.1705039171485, 380.0609558006913]
Principal point: [315.95206735330476, 246.03324000920372]
Distortion model: radtan
Distortion coefficients: [0.03381415466102244, -0.04224089860000907, -0.0015946682870833341, -0.0011350490472189934]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.02 [m]
Spacing 0.006 [m]

IMU configuration

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IMU0:

Model: calibrated
Update rate: 200.0

Accelerometer:

Noise density: 0.023109555248423664

Noise density (discrete): 0.32681846452731084

Random walk: 0.0006236165627200036

Gyroscope:

Noise density: 0.002586178988238998

Noise density (discrete): 0.036574093998919206

Random walk: 1.5222428152660453e-05

T_{ib} (imu0 to imu0)

[[1. 0. 0. 0.]

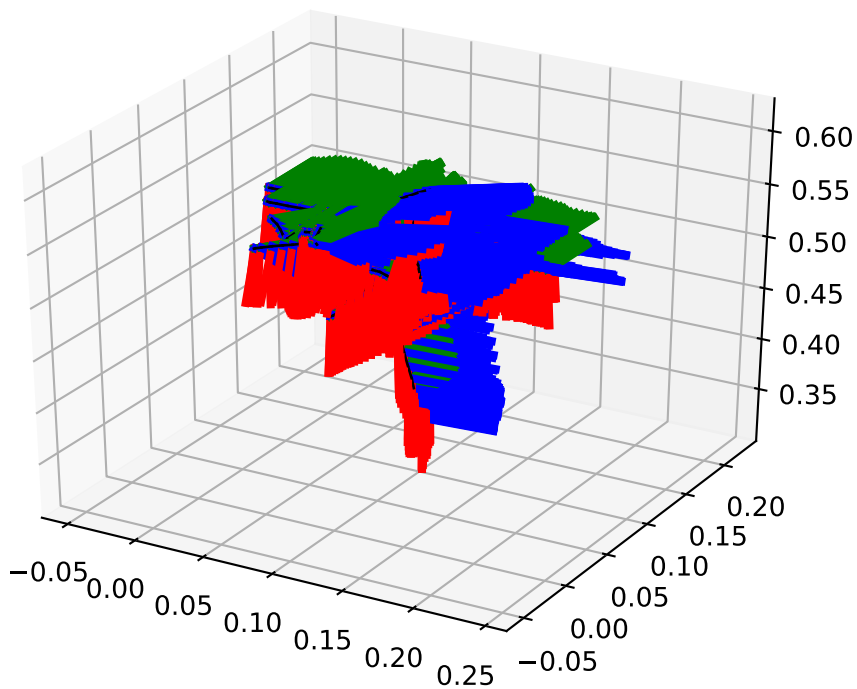
[0. 1. 0. 0.]

[0. 0. 1. 0.]

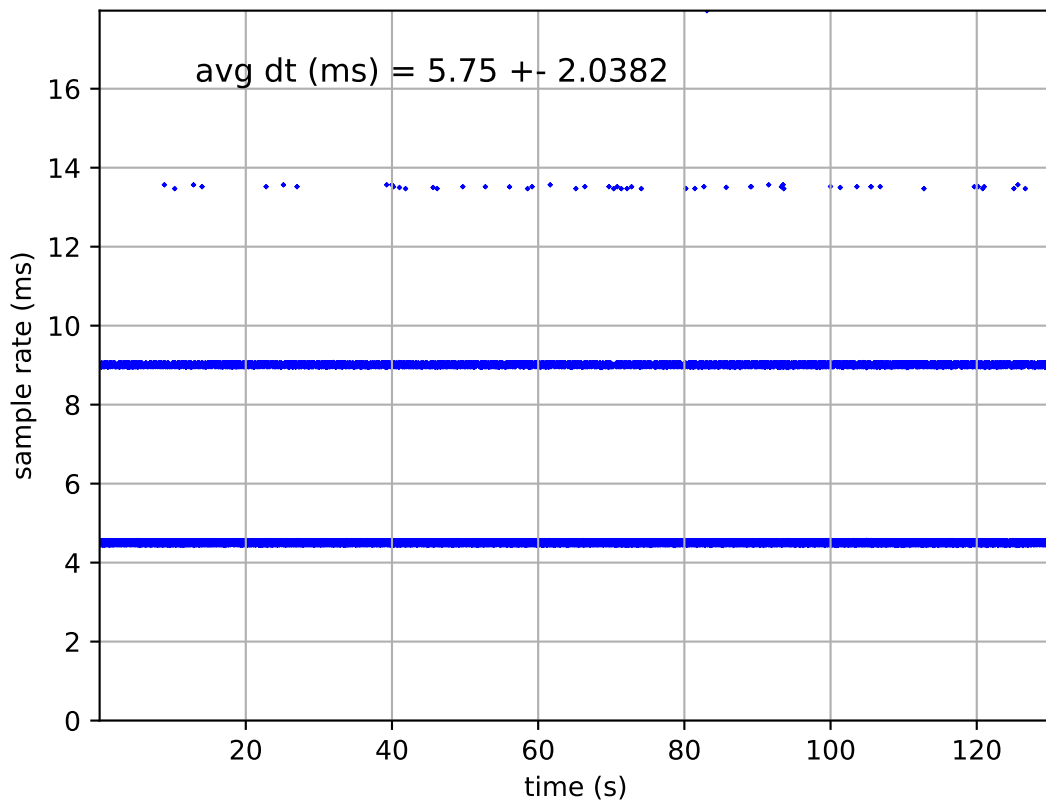
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

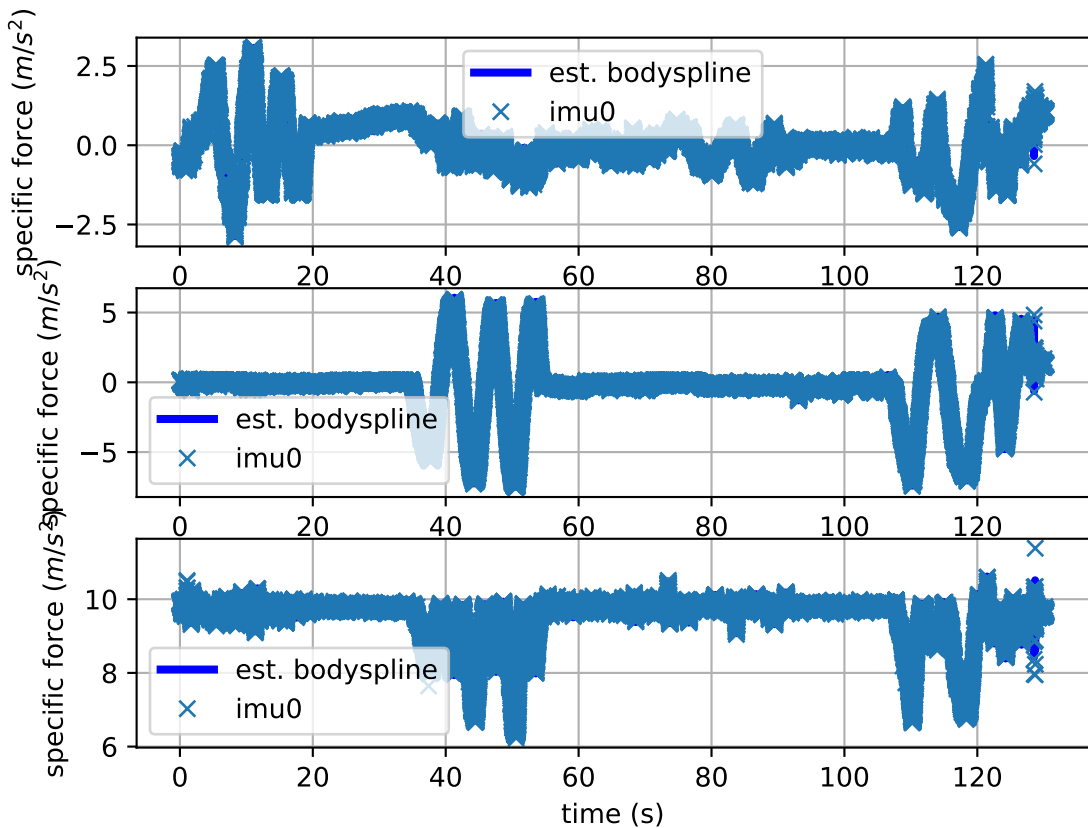
imu0: estimated poses



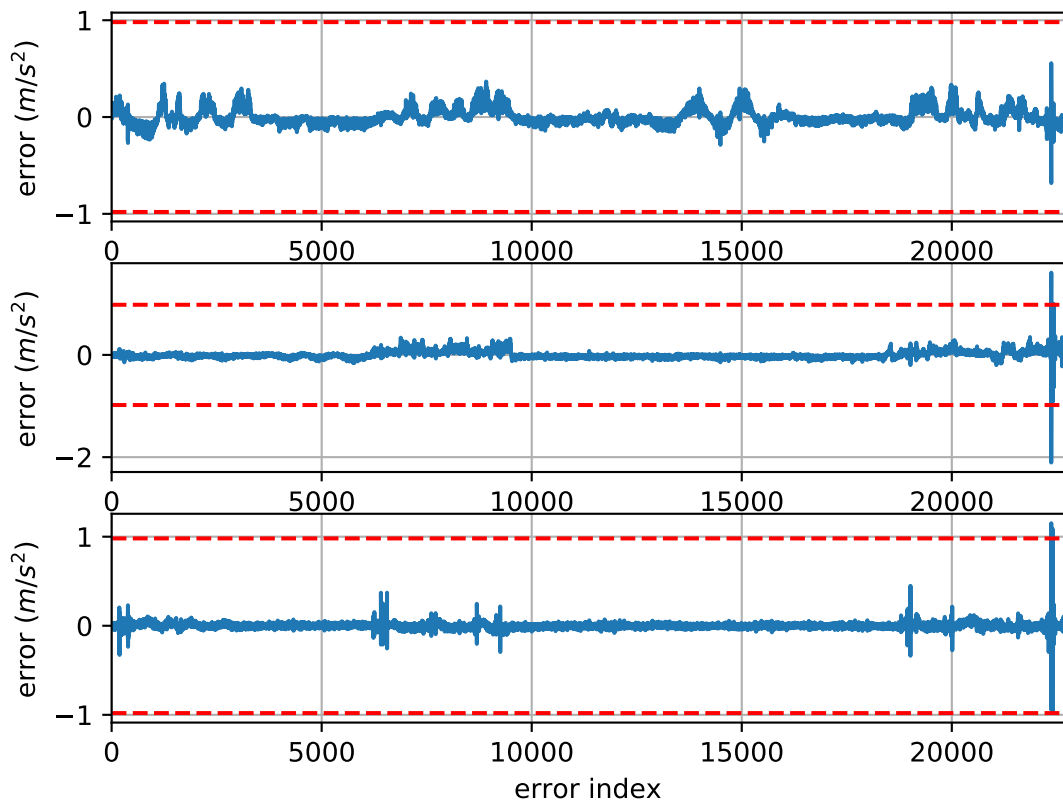
imu0: sample inertial rate



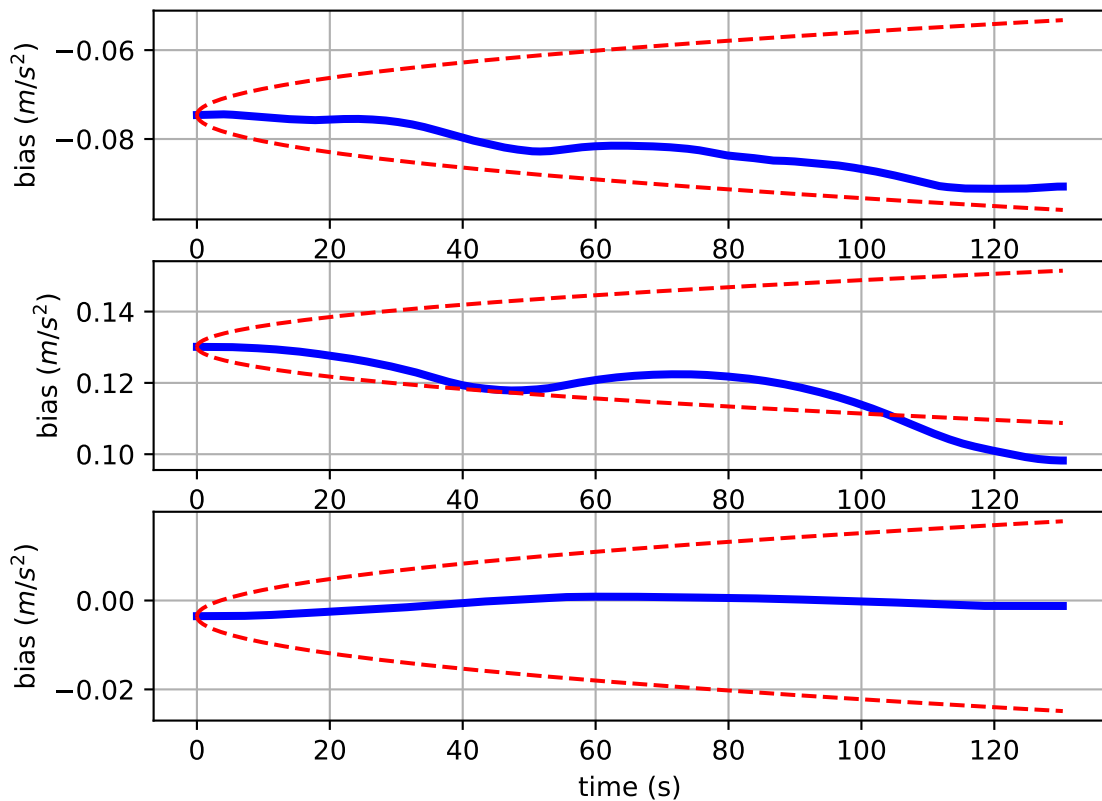
Comparison of predicted and measured specific force (imu0 frame)



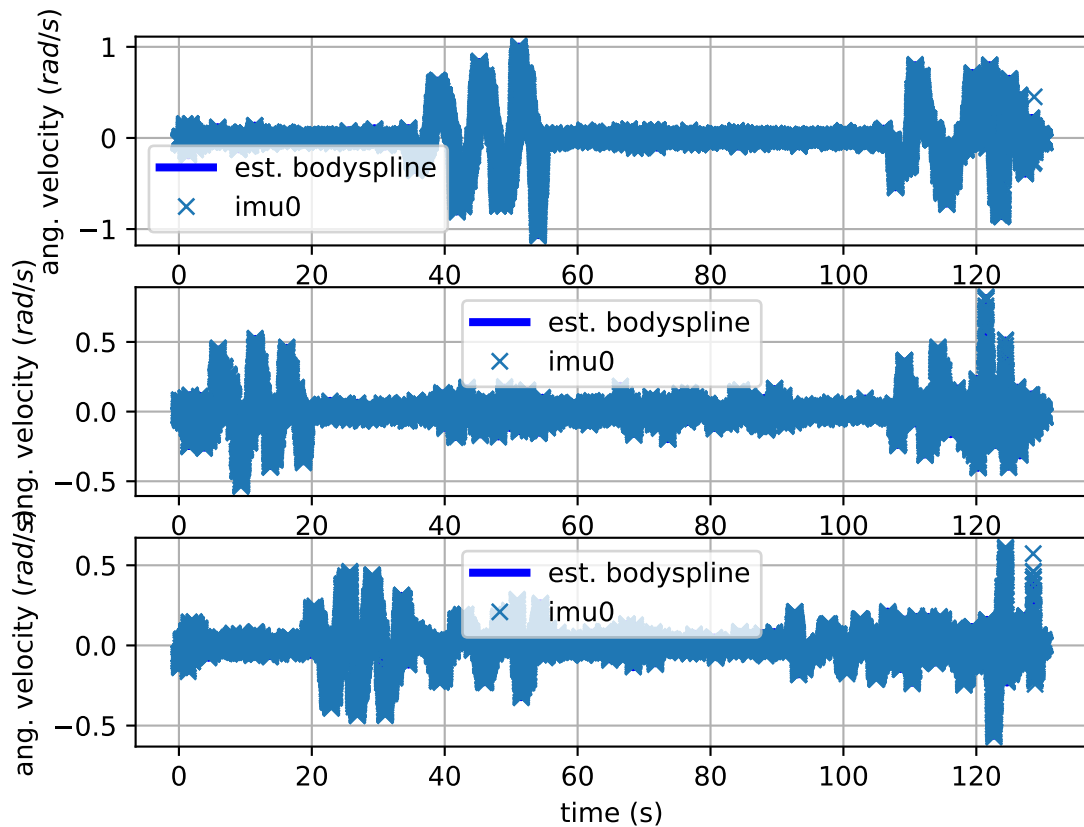
imu0: acceleration error



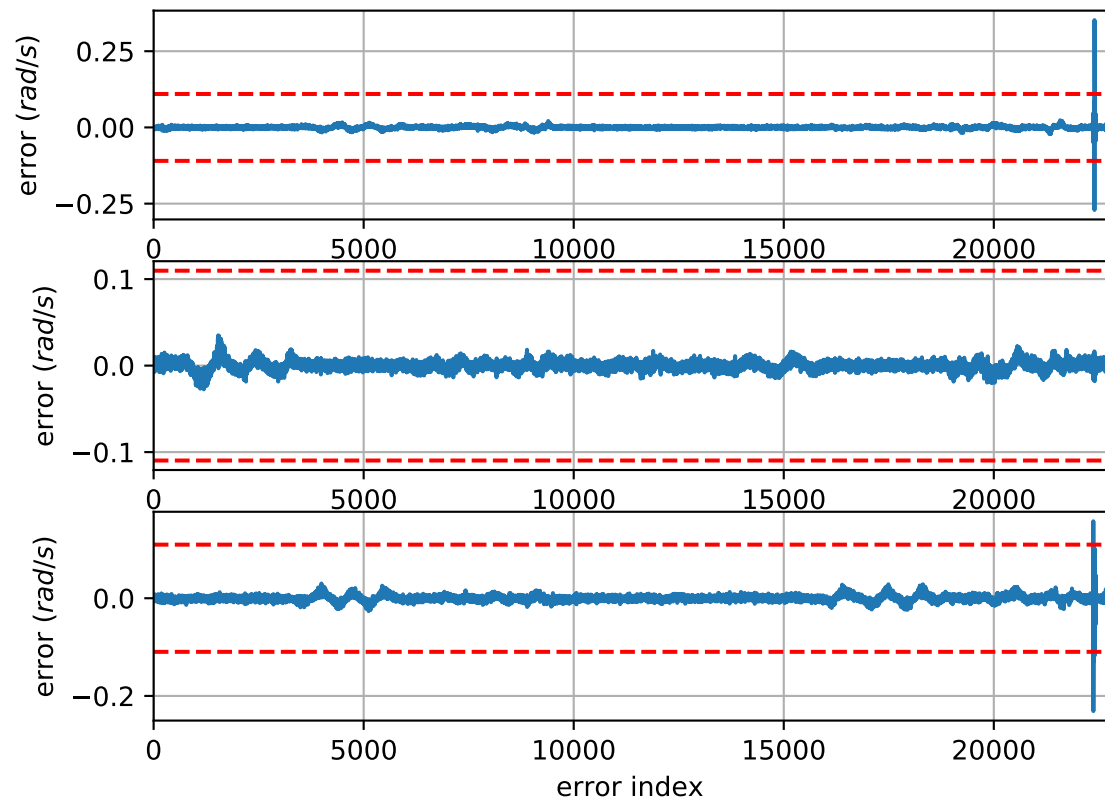
imu0: estimated accelerometer bias (imu frame)



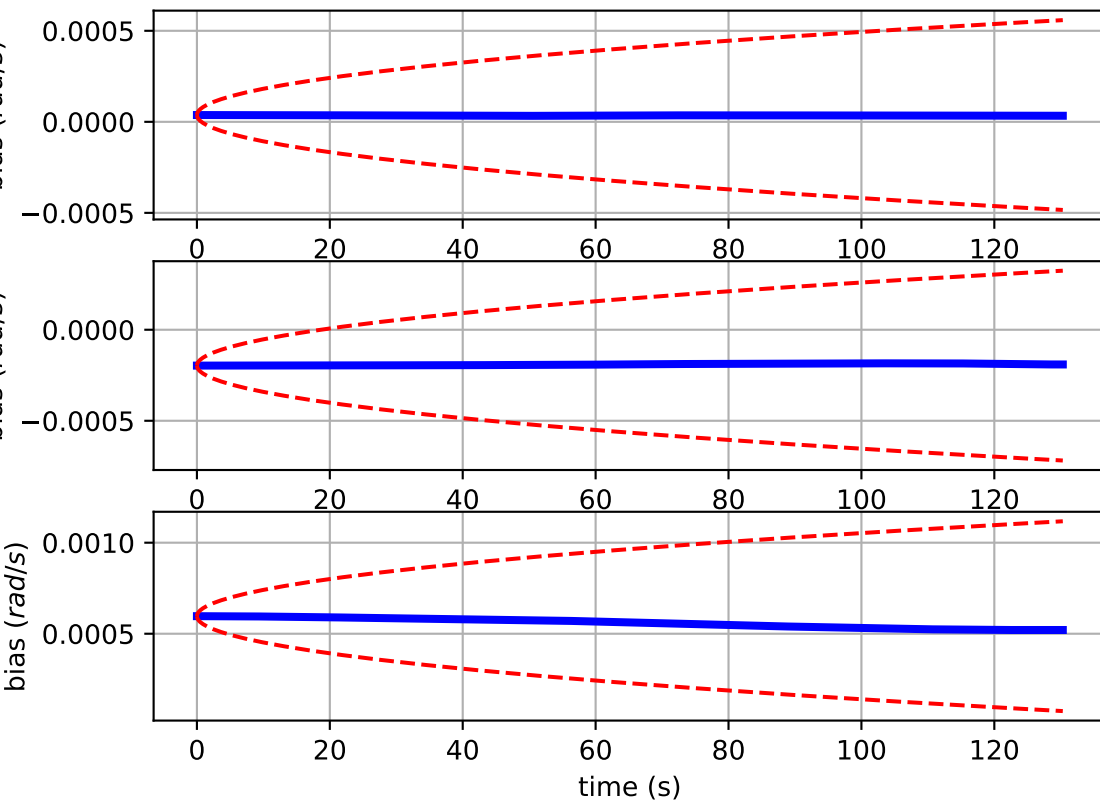
Comparison of predicted and measured angular velocities (body frame)



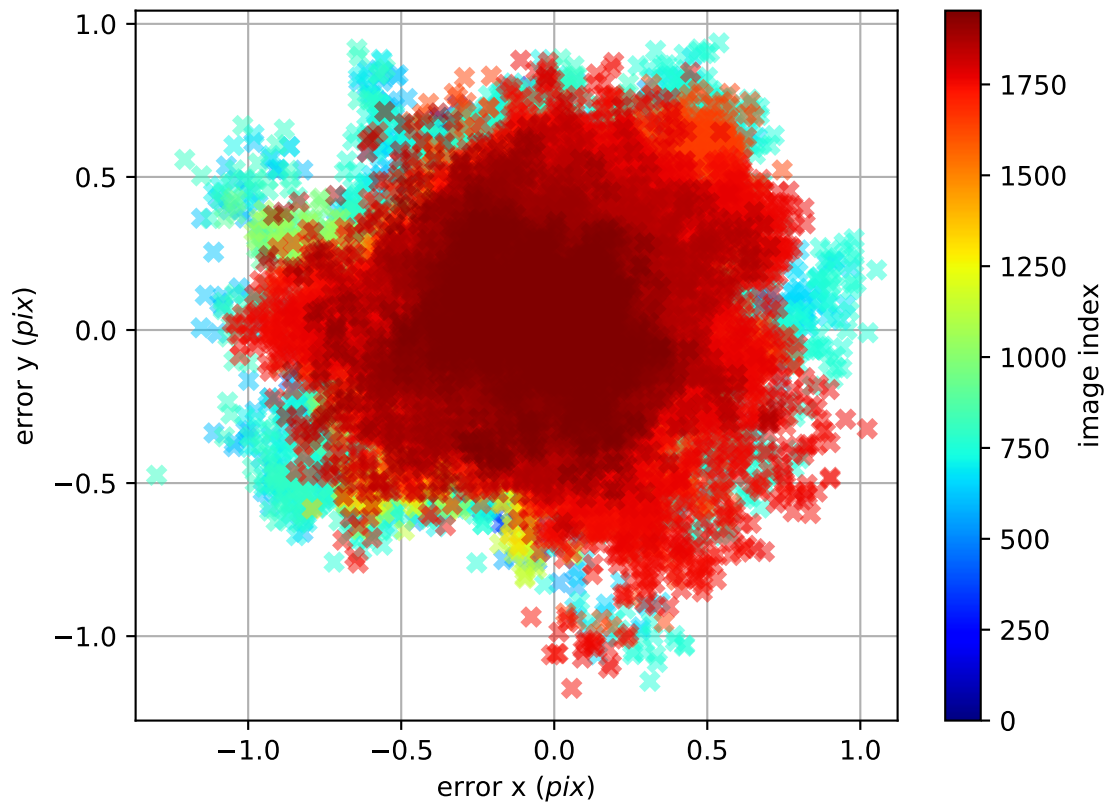
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

